Aston University
BIRMINGHAM UK

ME2502 Group Project

Group No. 6

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DFA: DFA principles were applied throughout the development of the battery holder to minimise assembly complexity, reduce the number of components, and improve reliability. The design evolved through multiple iterations, with particular focus on simplifying fastening methods, improving part alignment, and ensuring an intuitive assembly sequence.

During the first design stage, we identified the key requirements our design needed to meet and generated multiple concepts that met these criteria. Each one minimised material use, product cost and incorporated a quick assembly whilst keeping structural integrity. The concepts were then evaluated as a group based on assembly complexity, manufacturability, and stability, leading to refinement of the overall geometry and fastening strategy. (see Fig.1 For Concept Drawings)

We then identified key mechanisms and components in each that would work well together to make a final design. We settled on a final design that consists of two primary structural components, (a top housing and a bottom housing), with a simple push fit joining the 2. This significantly reduced assembly difficulty and handling requirements, making it ideal for mass production. Minimising the number of components reduces the likelihood of assembly errors and improves overall reliability while also reducing assembly time through needing less assembly robots in our production. (See Fig.2 for prototype 2 sketch and 3D render) From this in our assembly line we only have one assembly stage, done robotically due to the simplicity of our assembly process.

A snap fit fastening method was initially selected to minimise part count and enable assembly without standard components. However, prototyping revealed excessive interference between components, making the fit too tight, therefore resulting in breakage during assembly. The fastening method was redesigned as a result to a push fit connection, allowing easier assembly while supporting sufficient retention force and improving reliability. (See Fig.3 for failed prototype 2)

Chamfers of 45° were applied to external edges to remove sharp corners, improving safety and facilitating easier component alignment during assembly.

The battery holder was designed around cylindrical cells measuring approximately 70 mm in height and 19 mm in diameter. Recessed battery compartments with a diameter of approximately 19.5 mm allow batteries to be easily inserted while preventing excessive movement during operation. This removes the need for added retention features and simplifies assembly. (See Fig.4 for added grooves and bus bar)

Sliding slots were incorporated to allow easy insertion of the bus conductors. Groove features were included to prevent excessive movement of the bus bar and ensure consistent positioning during operation. This modular approach simplifies assembly and reduces the need for other fastening components.

To improve structural stability, we added an M5 thread to go through both the top and bottom parts. They are fastened using nuts and washers, increasing overall mass, yet we decided improving rigidity was vital. The method enhances stability without adding significant assembly complexity, supporting the simplicity of the push-fit design. The design underwent further refinement to improve stability during assembly. An added thin support disc with battery holes was introduced above the recessed compartments to prevent cells from tipping during handling, improving assembly control without significantly increasing complexity or weight. It reduced assembly time from 50 seconds to ≈ 30 seconds. The push-fit connection was also adjusted to provide a slightly tighter engagement after initial testing indicated excessive looseness, improving retention and structural stability. Additionally, a central hole was incorporated to allow the option of a thread and nut connection as a contingency for increased robustness if needed. These modifications demonstrate iterative improvement based on testing feedback while maintaining the simplicity and performance of the overall assembly. (See Fig.5 for final CAD render)

Overall, the final design shows effective application of DFA principles through part count reduction, simplified fastening methods, self-finding features, and logical assembly sequencing, resulting in improved assembly efficiency and reliability.

DFM:

DFM focuses on improving product design to enable efficient and cost-effective production while maintaining performance. DFM principles were applied to optimise material choice, geometry, tolerances, and manufacturing processes throughout development.

We selected ABS plastic as the primary material due to its good impact resistance, ease of manufacturing, and suitability for additive manufacturing. The material also provides electrical insulation and structural strength for the application, making it well suited for prototyping and potential production also making it ideal for providing a safe and secure structure for the batteries. The prototype was produced using additive manufacturing through 3D printing, allowing rapid iteration and testing of various design configurations. This approach enabled quick refinement and testing of tolerances and fastening methods before finalising the design.

Additionally, the bus conductors were manufactured using water jet cutting, which is well suited for producing precise profiles without thermal distortion. This process maintains material properties while producing clean edges with minimal post-processing, ensuring consistent quality and efficient manufacturing. The battery slot diameter was refined to 19.5 mm to provide sufficient clearance for insertion while keeping stability. This tolerance adjustment improves assembly reliability and reduces the risk of interference during manufacturing, ensuring consistent fit across components. The base is a small circular structure that allows easy insertion of the batteries, reducing assembly and disassembly time while avoiding a huge increase in material use (only 4mm wide). Throughout the design we used simple geometries, reducing manufacturing difficulty and improving print reliability. This also reduces production time and improves repeatability, making mass production cost and time effective.

Rounded internal corners with a radius of 0.50 mm were used to reduce stress concentrations and improve manufacturability, which improves durability and ensures smoother material flow. On areas like the battery holes and the push fit, this also allowed for easier use and assembly, making for smooth insertion of both.

We used a wall thickness of 6 mm for structural strength while minimising material usage and print time. Maintaining consistent wall thickness improves print quality and reduces the likelihood of defects.

Bus conductor grooves were designed with a depth of 1 mm to ensure proper positioning and preventing excessive movement during operation. This depth was chosen with material reduction in mind, as only 1mm was needed to prevent the bus bar from moving. The main slots on top and bottom have widths of 7.5mm to allow easy access to the components and reduces material- this therefore reduces both product cost and manufacturing time.

The assembled battery holder has an overall diameter of 85 mm and a height of 76 mm, ensuring sufficient space for the batteries and bus plate to be tightly packed to support a compact structure. When designing the side supports, we found a balance between material reduction and weight support, shown through the design evolution. In prototype #1, we started with strut widths of 30mm and eventually ended with widths of 18.9mm in the final product. We removed excess material around the push fit, deciding it wasn’t necessary for the final product as it increased weight, cost and material without providing any benefit to the push fit.

When mass produced, our product has an estimated material cost of £0.25 for the ABS, £0.44 for the part production, £1.00 tooling cost, £0.12 for the bus bar, £1.25 robot assembly, giving us a total assembly cost of £5.10. The assembly line produces 48 units/hour, giving an annual capacity of approximately 96,000 units. Given the required production volume of 5,000 units/year, the system operates at 5.2% capacity utilisation.

We included chamfers of 45° throughout to remove sharp edges, improving safety and reducing stress concentrations during manufacturing while also enhancing part handling and ease of assembly

The design underwent multiple iterations where fastening methods and key dimensions were refined based on testing results. This process improved both manufacturability and assembly reliability, proving effective application of DFM principles. Overall, the final design balances manufacturability, structural performance, and production efficiency through material optimisation, simplified geometry, tolerance refinement, and iterative testing. Going forward to develop our design even further we could improve the quality of the push fit itself, ensuring it doesn't need a thread and bolt to keep it compressed together. This would reduce weight and costs and help the manufacturing become easier with less steps and parts to assemble. To achieve this, we could add a small lip in the middle of the side prongs on the CAD design itself, so it causes it to catch on in the slots when inserted. This would create enough force and pressure for the holder to compress the batteries and hold the weight of them by itself without needing the thread component.

Smart Manufacturing:

In figure 6 is our group's SMART manufacturing facility that we drew out and annotated in previous sessions. When prototyping, we used an FDM (Fused Deposition Modelling) 3D printer where the whole design of our battery holder is made. We chose this machine as it prints the parts on demand, meaning we could use it in our own time and only when necessary, reducing cost. It allowed easy analysis, testing and design improvement, making it a perfect choice when prototyping.
When mass producing the product, we will use injection moulding. Due to the production quantity needed, injection moulding has a fast cycle time with a low cost per unit, making it perfect for producing high quantities of the same exact product. Material waste is minimal, and injection moulding delivers high dimensional accuracy with consistent wall thickness, giving excellent repeatability from part to part.
For the second step in assembly, we chose to have a humanoid robot rather than a traditional robotic arm as part of our conveyor belt as it can move to other workstations if necessary. This would be efficient for us as our metal bus plate will be die cast separately and then moved (by conveyor belt) to the workstation labelled as 3 in figure 6. This then allows the robot to collect the part made from the injection moulding, remove all the excess material, go to workstation 3 and collect the bus plate, and then eventually assemble all the parts together. This includes inserting an M5 thread and fastening on nuts/washers at either end, something which we added in the final design in the past couple of weeks. As these are standard components, they are also cheap and easy to incorporate in the manufacturing process, as the humanoid robot wouldn't face any issues in assembling the product. After all this, the final products should come to the end of the conveyor belt where a human is there to check, test and pack the battery holders ready to be shipped off to a client. This final step of quality control eliminates any potential errors in production and removes any undetected errors in the robotic assembly line.
Throughout the manufacturing we would use a large range of software and hardware to make sure the quality is maintained for each of our battery holders. This would include:

- Thermal Cameras- used to control quality of the product and ensure the temperature of the battery

pack is stable and will prevent the overheating and deformation of materials

- Filament Sensors- used to measure and prevent any filament breakages
- GOM inspect- used to scan products and compare them to CAD, giving reports if anything is incorrect from drawings and diagrams
- Profilometers- measures surface finish, allowing efficient time management as surfaces don't need to be sanded each time.

With visiting the GBS IoT factory last week, we will use the cameras and software listed above both in a separate digital quality control stage as well as the human quality control stage as part of our SMART manufacturing facility. This would greatly reduce the risk of errors and allows the product to be produced at a much higher standard.
This SMART facility directly supports and includes Industry 4.0 ideologies as it is interconnective, (as machines can continuously share data), it has autonomy, (as there is minimal human intervention and flexibility), and the product can be produced according to what is needed. (See Fig.6 for SMART facility design) Our product does all of this to a high output rate and quality, meaning it includes these ideologies in an optimal way.

Appendix:

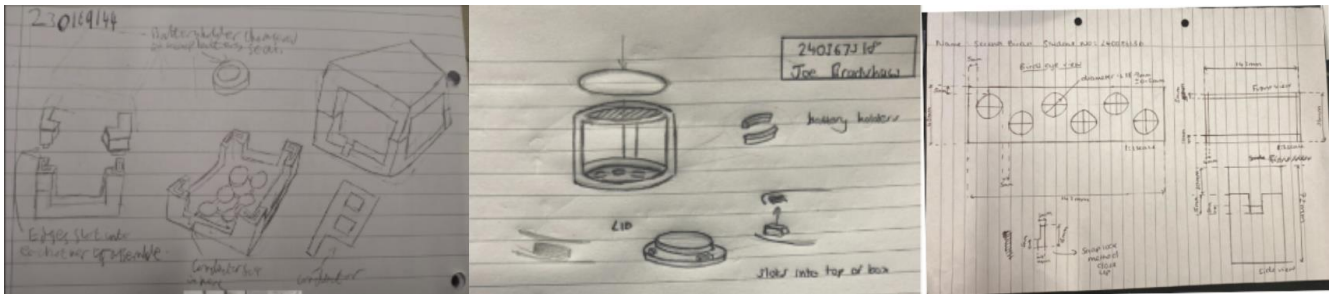


Figure 1: Concept Drawings



Figure 2: Prototype 2 sketch and 3D render

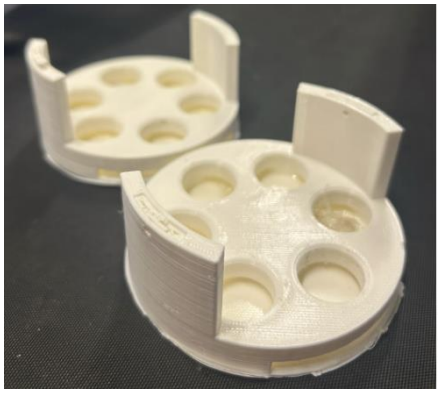


Figure 3: Failed prototype 2

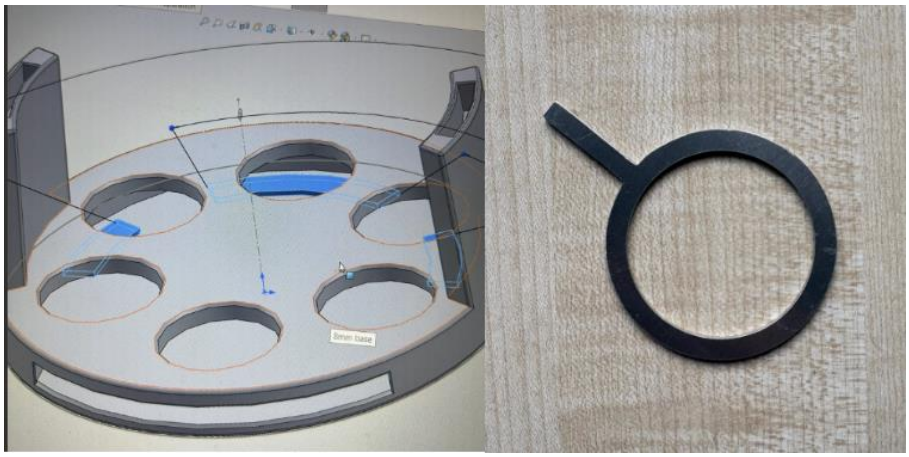


Figure 4: Added grooves and bus bar

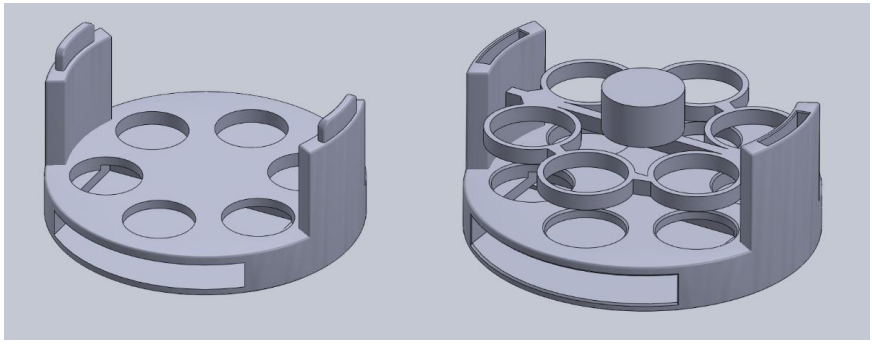


Figure 5: Final CAD render

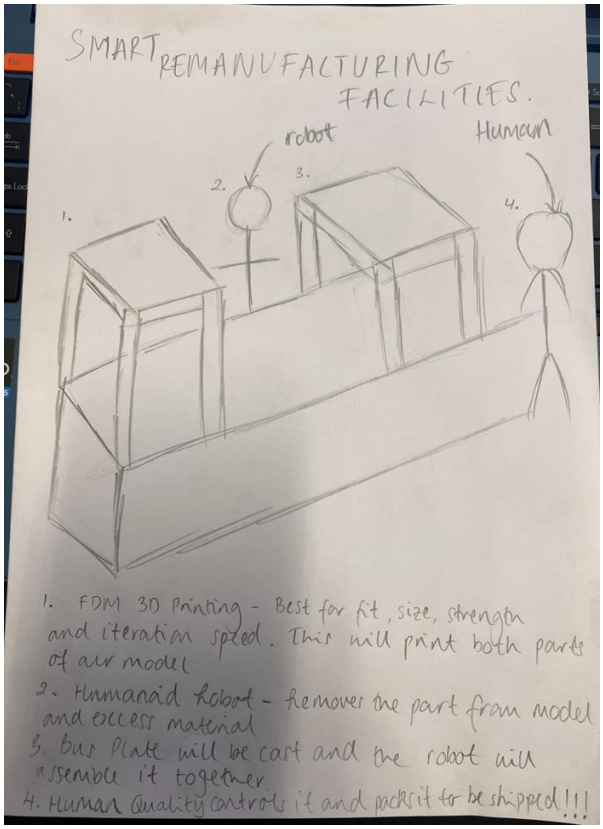


Figure 6: SMART facility